Santiago Neira Lopez

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Professional Summary

PhD student in Mathematics specializing in statistics, machine learning, and scientific computing. Experienced developing probabilistic models, deep learning methods, and numerical algorithms for large-scale scientific applications. Interested in Machine Learning, Data Science, and Quantitative Research internships.

Technical Skills

Programming: Python, R
Libraries: PyTorch, NumPy, SciPy, Pandas, GPy, Emukit
Machine Learning: Linear Models, Gaussian Processes, Bayesian Optimization, Deep Learning, Operator Learning, Generative Models, Probabilistic Machine Learning
Statistics: Statistical Inference, Bayesian Statistics, Probability Theory, Stochastic Processes
Applied Mathematics: Applied Mathematics: Numerical Methods, Optimization, PDEs.

Experience

Graduate Teaching Assistant

University of Massachusetts Amherst, 2023-Present.

- Led weekly discussion sections for 30–40 undergraduate students in Calculus I and II.
- Designed problem-solving sessions and provided one-on-one support, improving students' understanding of core mathematical concepts.
- Communicated complex mathematical ideas to students with diverse backgrounds.

Projects

Heston Stochastic Volatility Model (Collaborative Project)

- Implemented multiple European option pricing algorithms in Python.
- Built a calibration pipeline using global and local optimization techniques.
- Applied the calibration pipeline to real TSLA options data.
- Compared pricing accuracy and computational efficiency across numerical methods.
- Technologies: Python, NumPy, SciPy, Pandas
- Repository: GitHub

Education

PhD in Mathematics

University of Massachusetts Amherst, 2023–Present.

- Research Interest: Machine Learning, Statistics, Surrogate Modeling

MSc in Mathematics

University of Massachusetts Amherst, MA USA, 2023–2025.

BSc in Mathematics

Pontificia Universidad Javeriana, Bogota Colombia, 2019–2023.

Selected Courses

- Machine Learning
- Bayesian Statistics
- Statistical Inference
- Probability
- Stochastic Processes
- Numerical methods
- Mathematics of Generative AI

Honors & Awards

- 2023. Orden al Mérito Académico Javeriano
- 2023. Best GPA, Honors for Bachelor's Thesis